Last updated in September 2024

Devesh Kumar

Hyderabad | devesh1102@gmail.com | +91 9308716025 | Devesh Kumar | devesh1102

Education

Btech + Mtech **Indian Institute of Technology Bombay**, Electrical Engineering *June 16 – July 21*
- CGPA : 8.49/10 (Transcript)

Minor **Indian Institute of Technology Bombay**, Computer Science *June 17 – July 21*

Academic Achievements

- Ranked in top 1 percent among 2 lakh aspirants in JEE Advanced, 16 | Secured 99.1 percentile in JEE Mains, 16.
- Qualified for National Talent Search Exam (NTSE)

Experience

Microsoft India, Software Engineer Hyderabad, India

- **Azure Manufacturing** *April 2022 - Present*
  - Developed a Data manager microservice on Azure for Manufacturing Industries to manage the data generated by their IOT devices.
  - Worked on bicep templates which would enable automated resource deployment on azure. Implemented virtual networks and authentication between the resources. Ensured that resources used managed identity to improve security.
  - Worked on implementing new strategies Copilot pipelines. Increased the accuracy of copilot from 40 percent to 75 percent through RAG search of instructions
- **Azure Global** *July 21 – March 22*
  - Designed and Implemented backend for Data Sharing and Validation pipelines for Azure PAAS: Enabled customer to validate the data being shared with another customer via CDM schemas and enforce validation policies for data.
  - Implemented long running async jobs for our service to be able to run and poll pipelines with regular status updates. Implemented clients to interact with other Azure Services like: Synapse, Data Share, Storage Account, Microsoft Graph

Chicago Mercantile Exchange (CME) Group, Intern Bangalore, India

- **Quant Research Team** *May 19 – Jul 19*
  - Movements in interest rate curve: Identified the largest percentile movement of interest rate curve for given strategies using principal component analysis of the interest rate curves. This helps in testing the existing risk models so that they could perform better on extreme market events in the future
  - Margin offset analysis: Developed R tool to calculate the margin offset at the currency level across currencies for all accounts

Projects

Face Recognition Autumn 18
- Developed a facial recognition program on Matlab based on fisher discriminant analysis and principal component analysis, did a comparison between them (using cropped Yale dataset). Achieved 88 percent accuracy
- The program is capable of handling lighting and facial variation between the dataset and the input.

Syntactic sentence parsing using Recursive Neural Networks Autumn 18
- Applied an RNN to build a parsed tree-structure based on the phrasal category prediction of the words.
- Converted the Penn Treebank dataset to a binary form by chomsky normal form and unary collapsing.
- Reduced the syntactic phrasal tags to 6 subcategories and used pre-trained word embeddings.

Technologies

Languages: C++, C, C#, SQL, Bicep, Python

Technologies: .NET, Microsoft SQL Server, Azure Open AI, Kafka, NoSQL, Redis, Kubernetes, Azure Keyvaults, Azure Functions